



Reinforcement Learning in Agent-Based Models: Towards Robust Policy Design

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Phd Course in (Multi-Agent) Reinforcement Learning : a primer, with applications in economics

Sapienza University of Rome



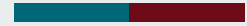


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Motivation

Designing robust policies with ABMs is **difficult**: requires exploring **many scenarios** and produces **static policies**





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Our Goals:

- Use RL to design policies that are **adaptive** and **robust to changes in the dynamics**
- Extend this framework to a setting where **multiple agents learn**





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Economic Focus: RL + macroeconomic ABMs → discover adaptive fiscal strategies that remain effective under uncertainty.





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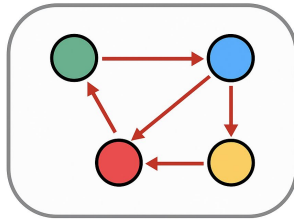
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Agent-based Models (ABM)

- **Definition:** Computational **simulations** of a system built from the bottom-up, where heterogeneous **agents** interact dynamically based on simple rules.
- **Objective:** To understand **emergent phenomena**

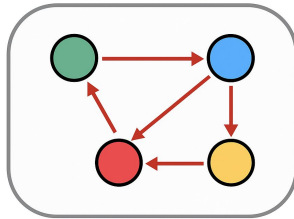
AGENT-BASED MODEL



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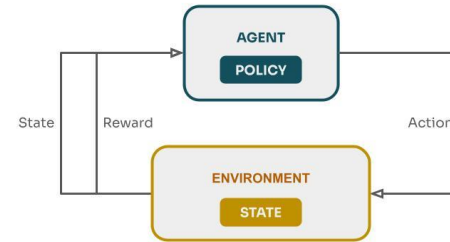
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AGENT-BASED MODEL



Reinforcement Learning (RL)

- **Definition:** Area of Machine Learning where an **agent** learns an optimal **policy** (behavior) through interaction with an **environment**.
- **Objective:** Maximizing a cumulative **reward** signal



Proximal Policy Optimization (PPO)

PPO is a state-of-the-art RL algorithm:

- **Stability:** Employs a **clipping mechanism** to restrict policy updates.
- **Exploration:** Learns a **stochastic policy** → better for complex exploration and multi-agent scenarios.
- **Flexibility:** Easily handles **continuous and discrete actions**

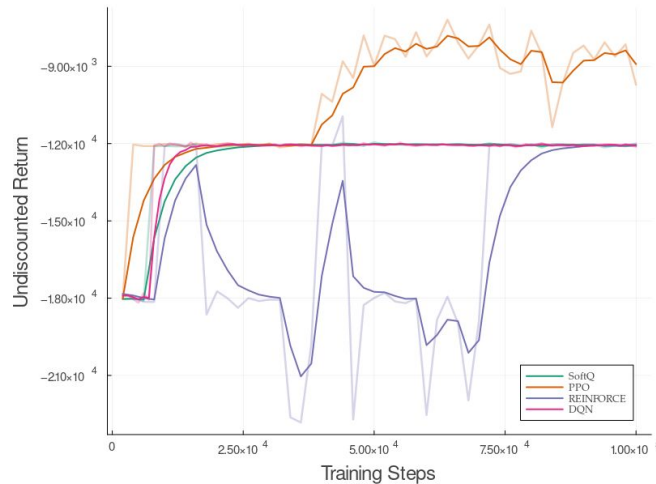




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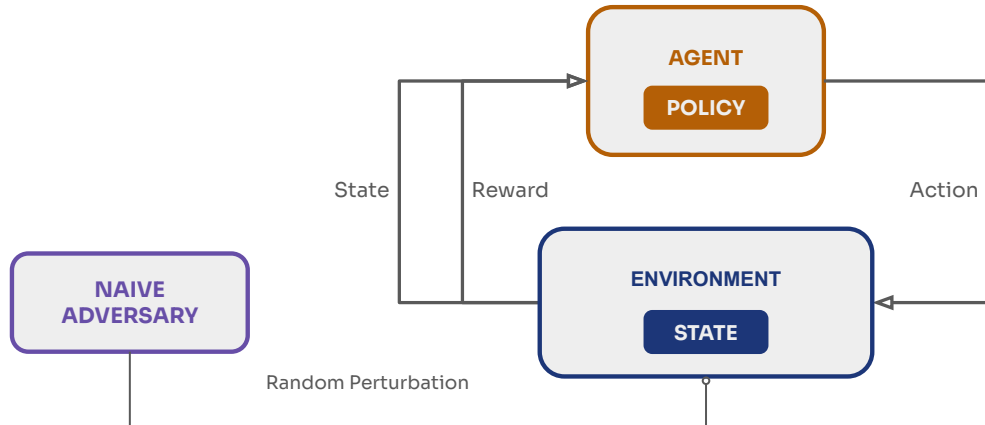
Strategy for Robustness: Treat Environment as an Adversary

Naive Adversarial Training (Domain Randomization)

- Adversary → **Stochastic**
- Randomly perturb environment parameters during training

Robust Adversarial Reinforcement Learning (RARL)

- Adversary → Trained to minimize the agent's performance
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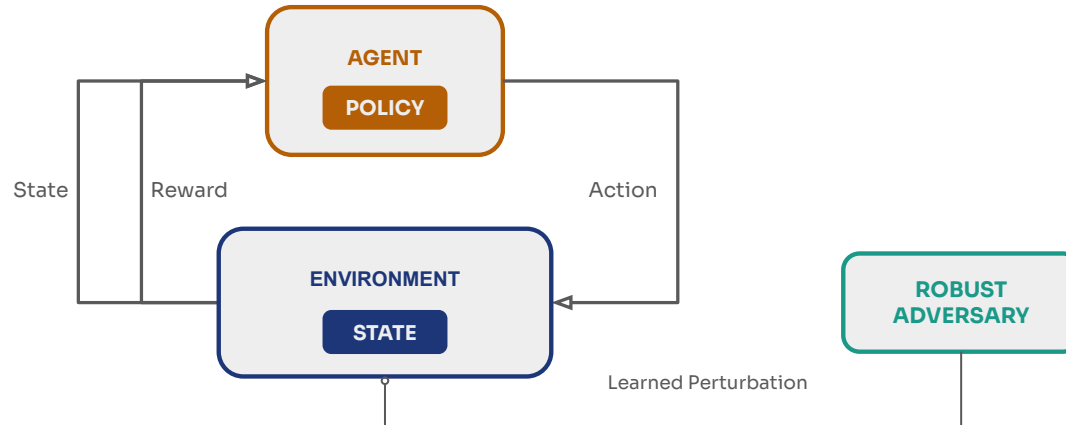




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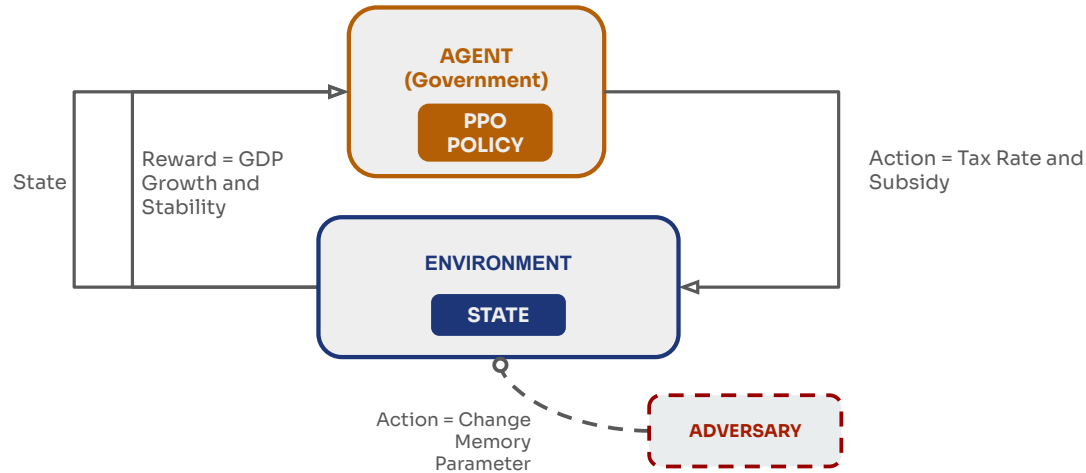
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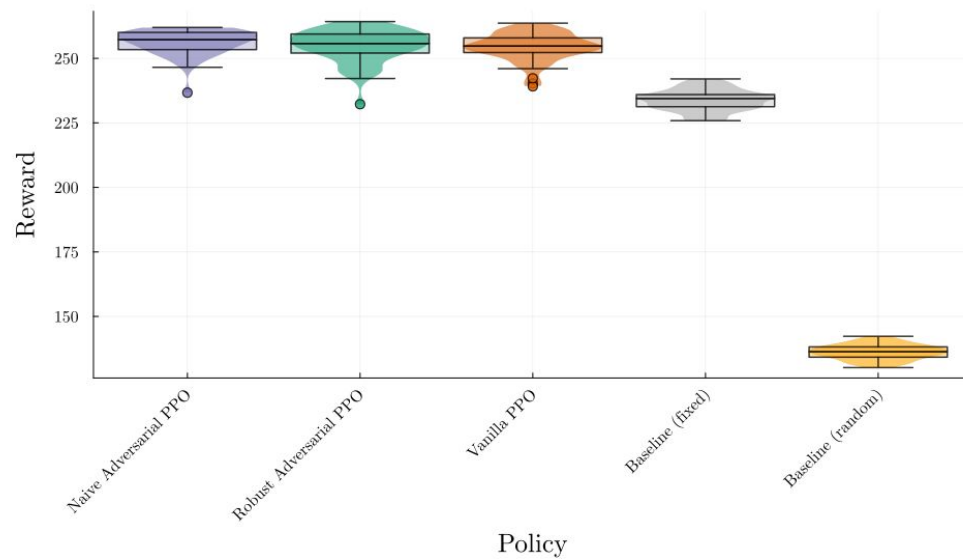
ABCredit

Medium-scale Macroeconomic ABM

- **Agent Actions:** control tax rate and unemployment benefit
- **Reward:** Balance GDP growth with economic stability
- **Adversary Action:** Change *Memory* parameter



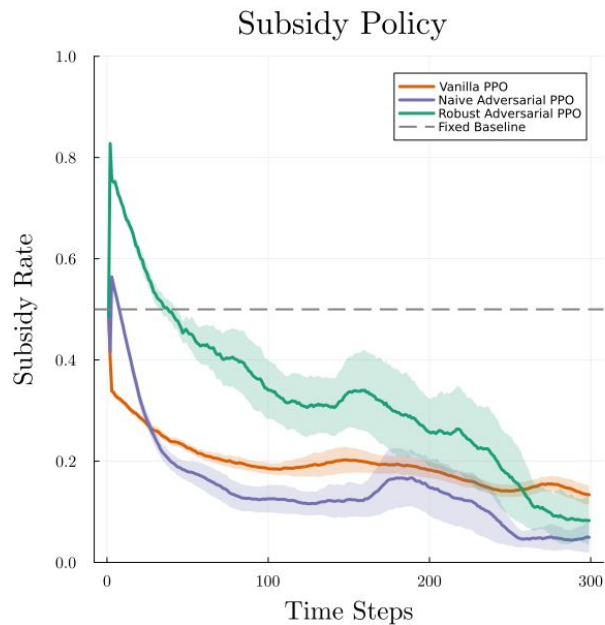
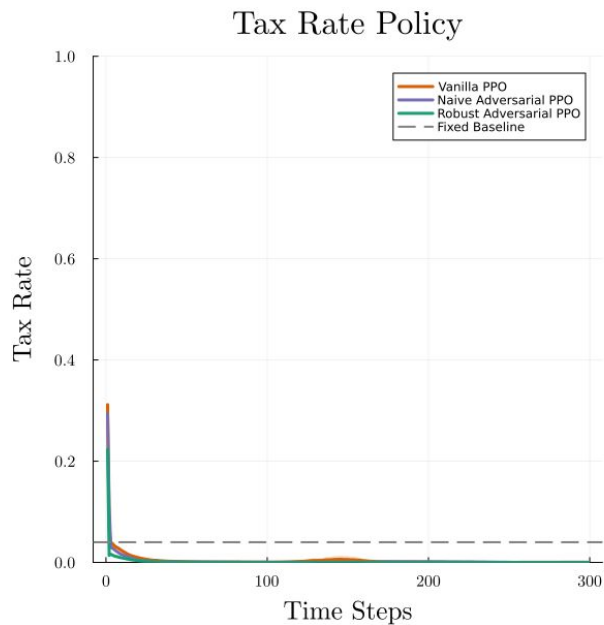
ABCCredit - Results



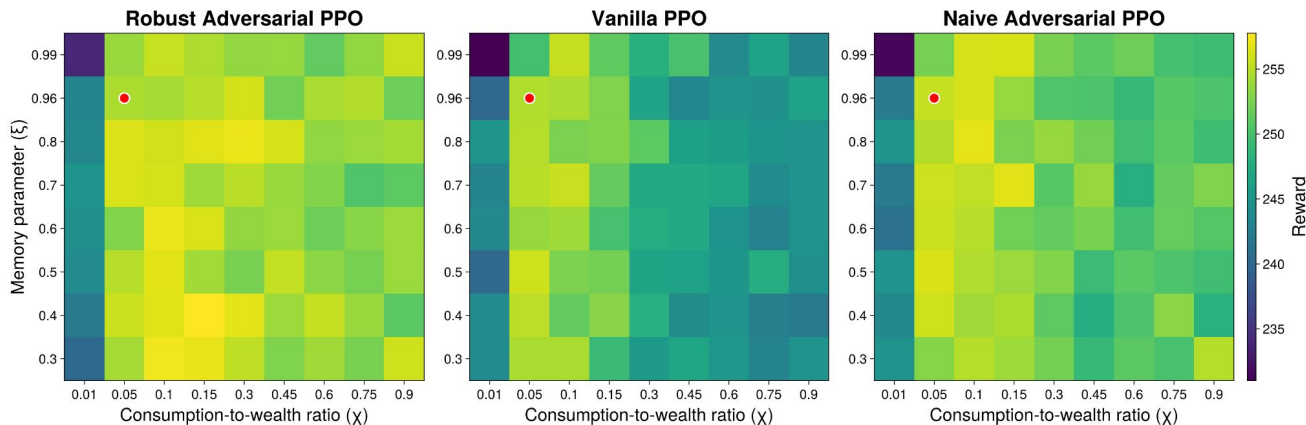
All RL policies perform better than the fixed baseline



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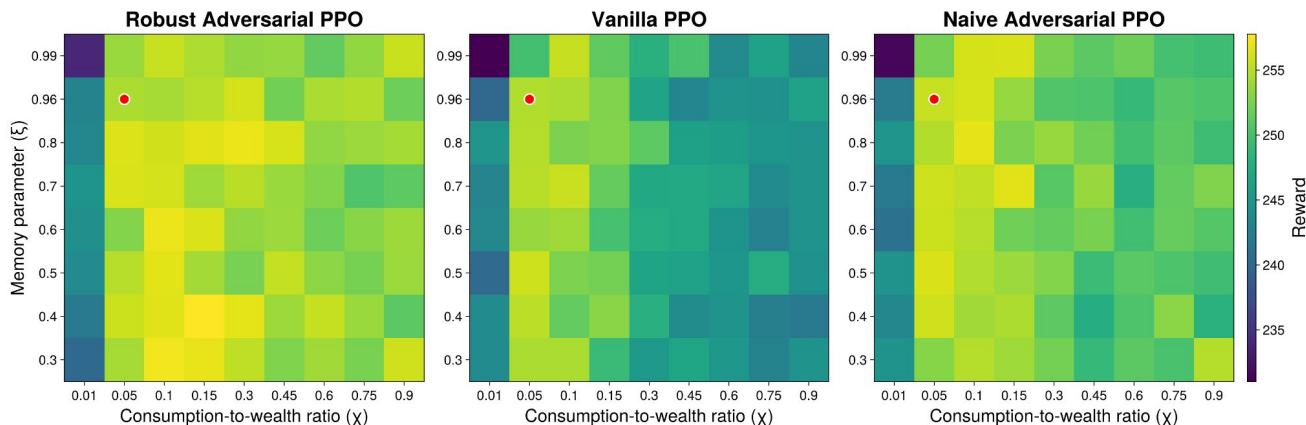
ABCCredit - Results



- Heatmaps show **average reward** across joint parameter space (y-axis: **perturbed** parameter, x-axis: **unperturbed** parameter)



ABCCredit - Results



- Heatmaps show **average reward** across joint parameter space (y-axis: **perturbed** parameter, x-axis: **unperturbed** parameter)
- **Cross-Parameter Generalization** : Adversarially trained policy generalize better also under shifts of parameters *unperturbed* during training





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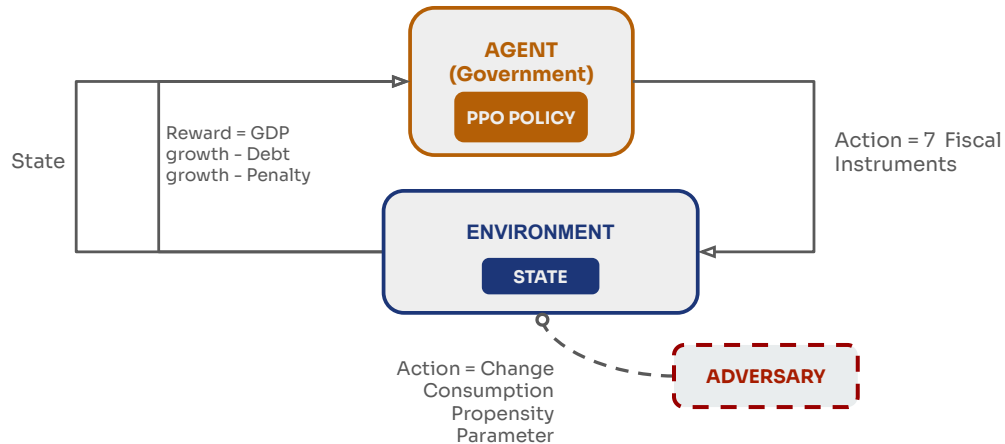
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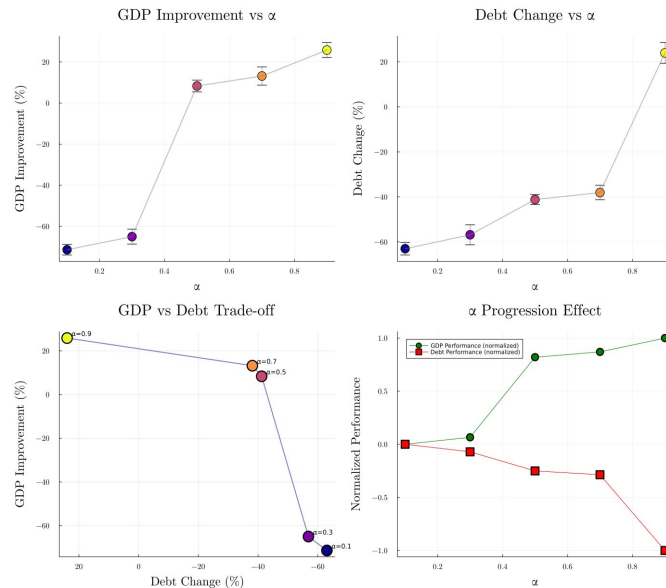
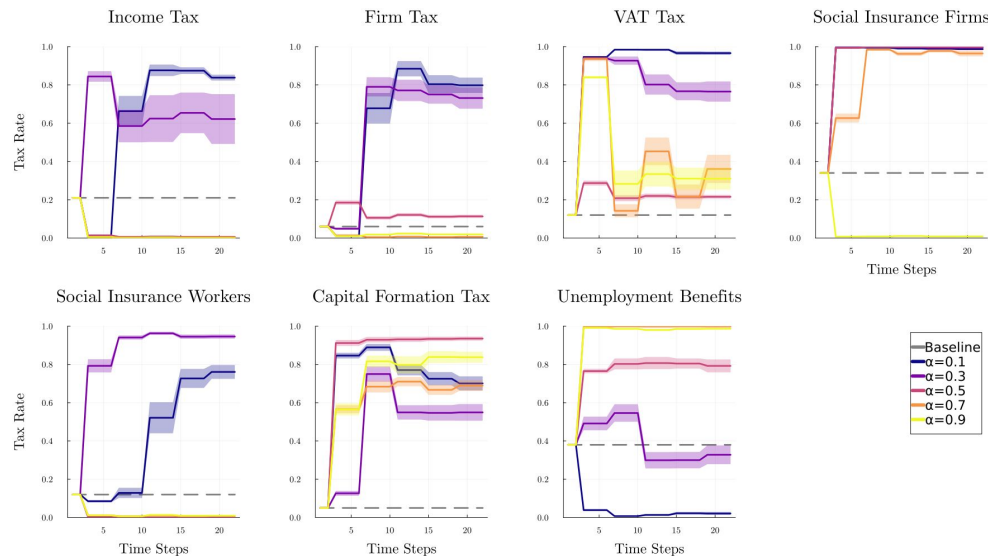
BeforeIT

Large-scale Macroeconomic ABM

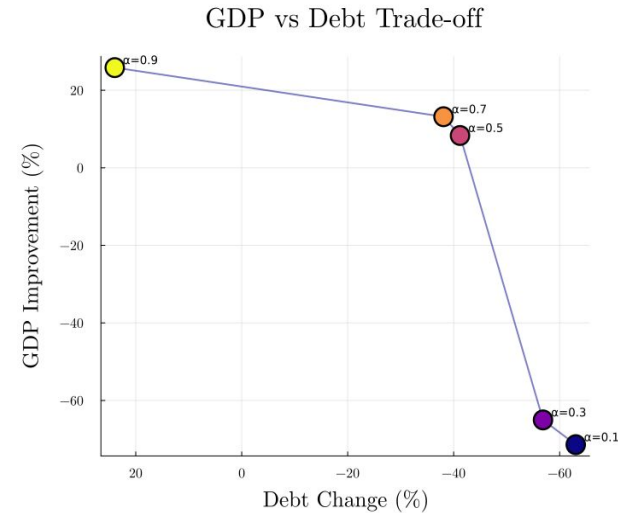
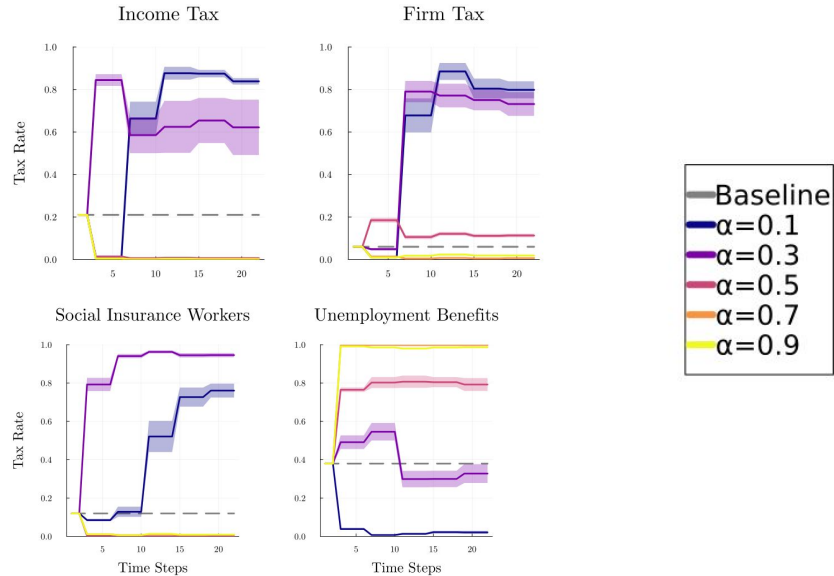
- **Agent Action:** control 7 fiscal instruments, updated every 4 timesteps (1 year)
- **Reward:** $\alpha \times \text{GDP growth} - (1 - \alpha) \times \text{Debt growth} - \text{Penalty for large tax changes}$
 - α balances the trade-off between growth and debt control
- **Adversary Action:** Change *Consumption Propensity* parameter



BeforeIT - Results



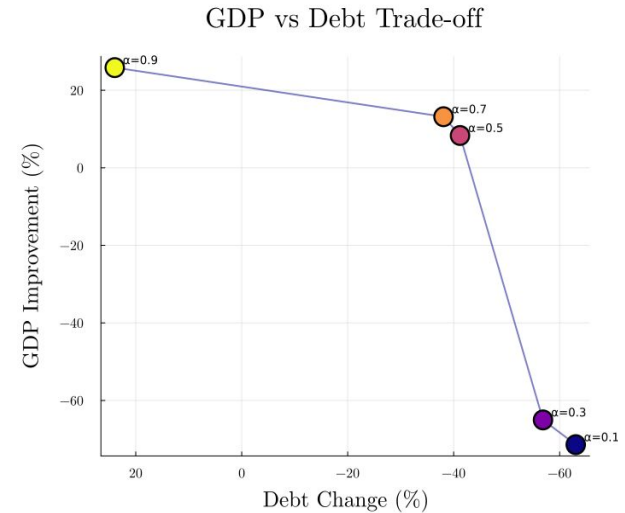
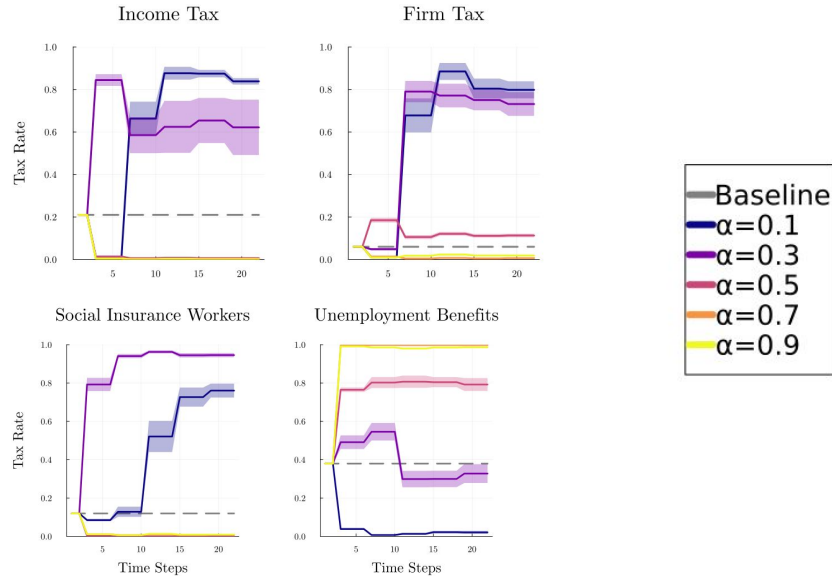
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- **Growth-focused policies ($\alpha=0.7, 0.9$):** Low taxes, High benefits → Higher GDP, Higher Debt



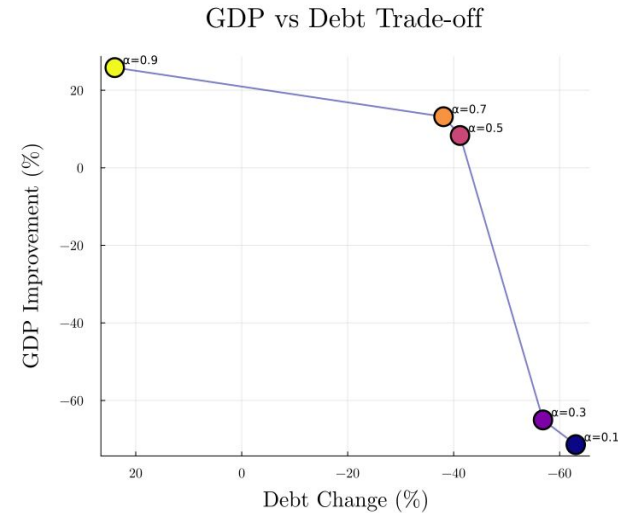
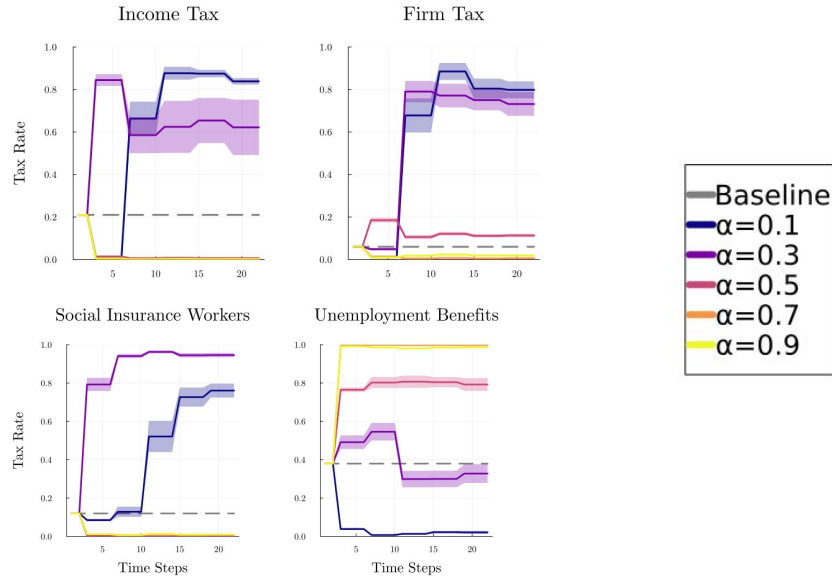
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- **Growth-focused policies ($\alpha=0.7, 0.9$):** Low taxes, High benefits $\rightarrow$ Higher GDP, Higher Debt
- **Debt-focused policies ($\alpha=0.1, 0.3$):** High taxes, Low benefits $\rightarrow$ Lower GDP, Lower Debt



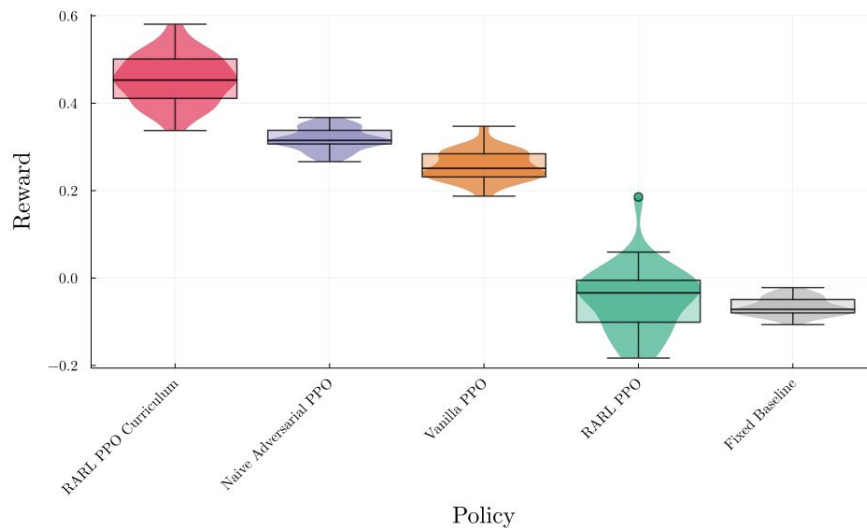
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- **Growth-focused policies ($\alpha=0.7, 0.9$):** Low taxes, High benefits → Higher GDP, Higher Debt
- **Debt-focused policies ($\alpha=0.1, 0.3$):** High taxes, Low benefits → Lower GDP, Lower Debt
- **Balanced policy ($\alpha=0.5$):** Good trade-off

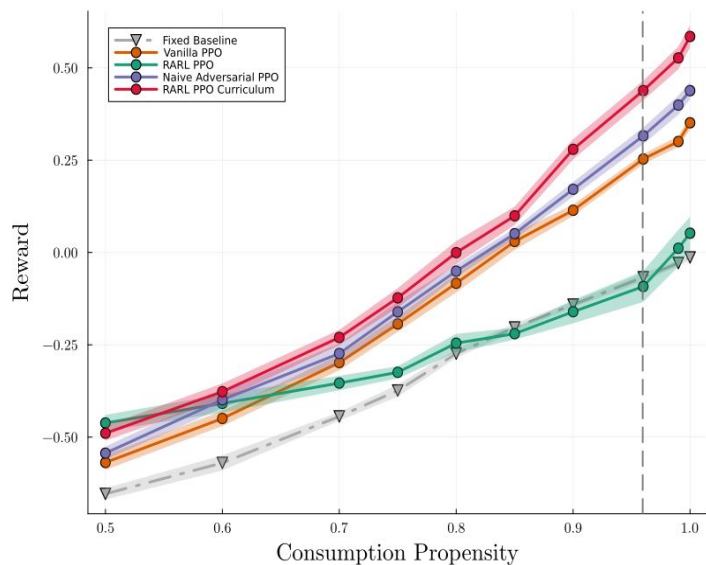


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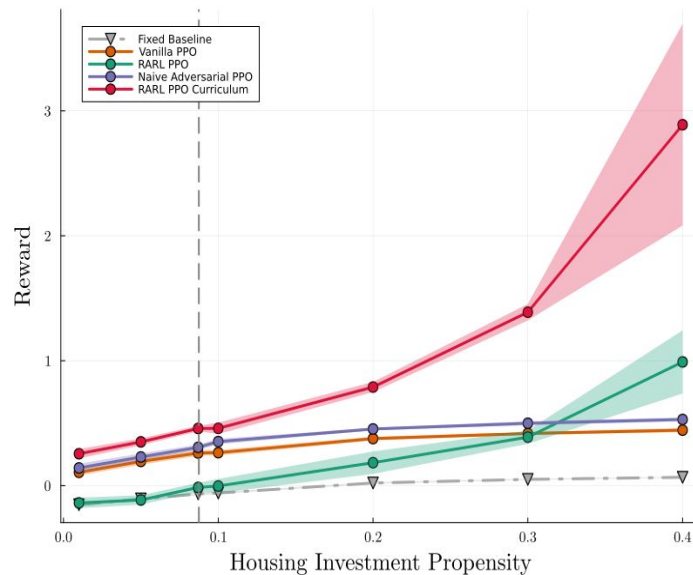
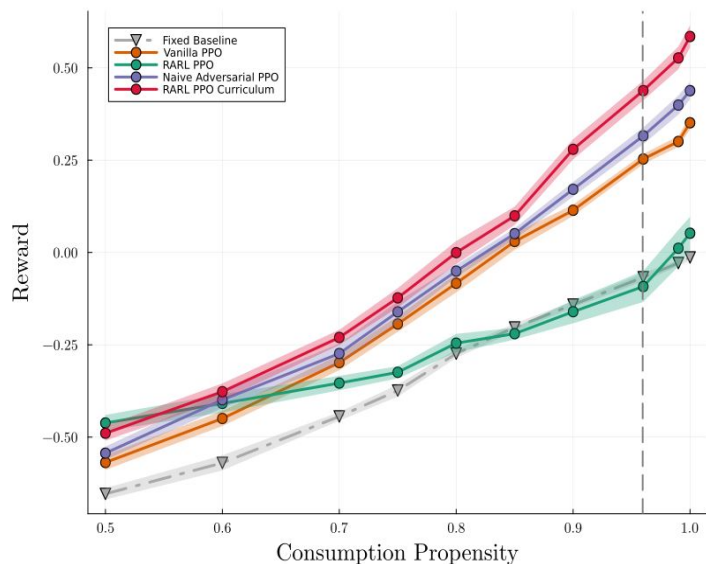


Curriculum Learning for RARL: gradually increase adversary strength over training

BeforeIT - Results

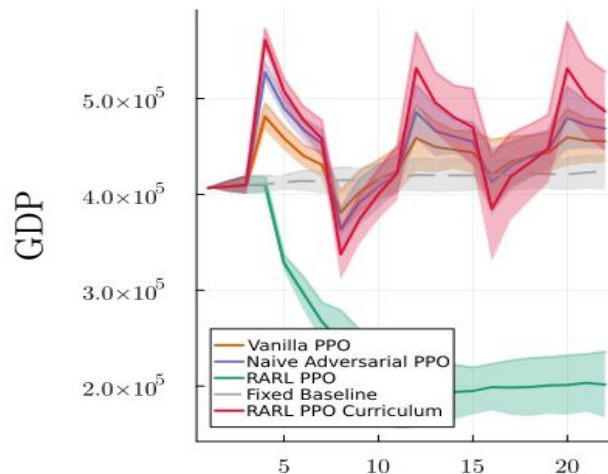
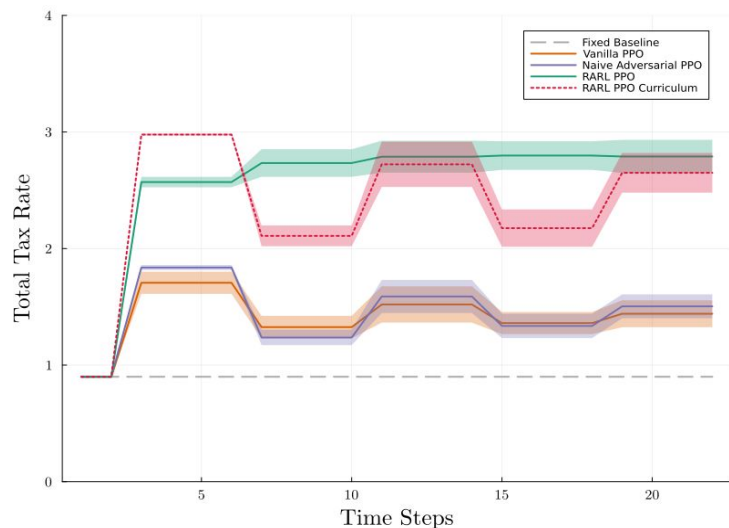


BeforeIT - Results



Successful adversarial policies are **more robust** to shifts of both **perturbed** (*left*) and **unperturbed** (*right*) parameters → **cross-parameter generalization**

BeforeIT - Results



Emergent macroeconomic principle → Robust policy learns **counter-cyclical fiscal policy**: raises taxes during booms and lowers them during downturns.





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Key Contributions

Adversarial RL Improves Robustness

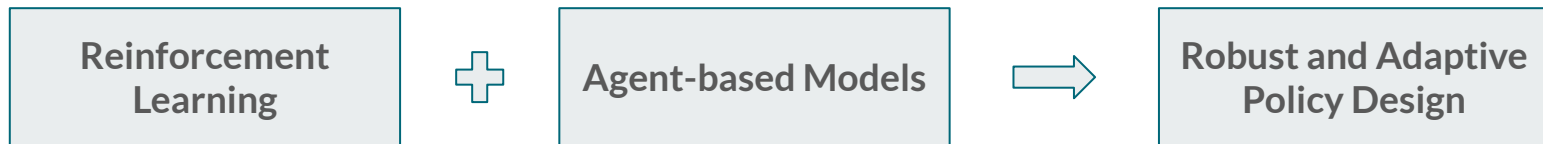
Curriculum Learning is Essential in Complex ABMs

Emergent Economic Behavior





Conclusions & Future Work



Limitations & Future Directions:

- Extend MARL to heterogeneous agents with competing objectives
- Improve reward function design for broader objectives (not just GDP/debt)
- Systematically choose which parameters to perturb





Thank you for the attention!

